



Deform

Bend a whole body, or selected faces, onto the profile with the same map -- a flat-modelled insert, core feature or tip protector.



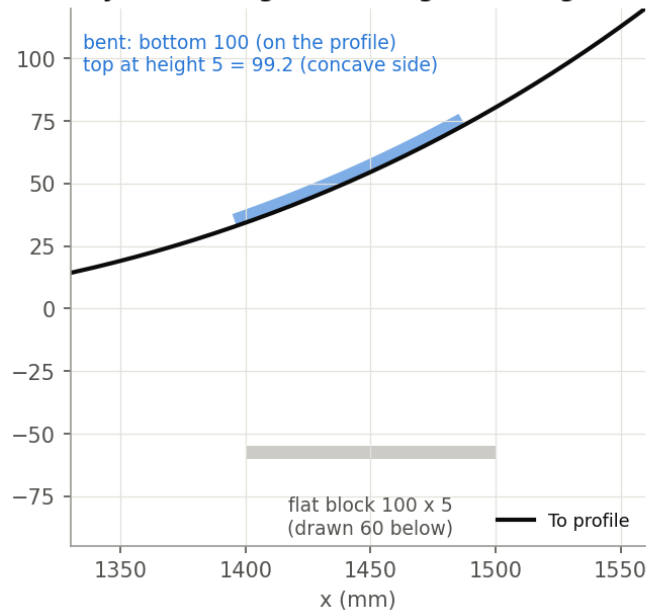
What it does

Parts modelled flat must sit on the curved ski.
Native Bend only flexes around an axis. Deform maps every vertex, edge and face through the bending map, so the part follows the profile exactly as a layer would.

Use it when

- An insert or protector modelled flat
- Features that must follow rocker / camber
- Faces rather than a whole body

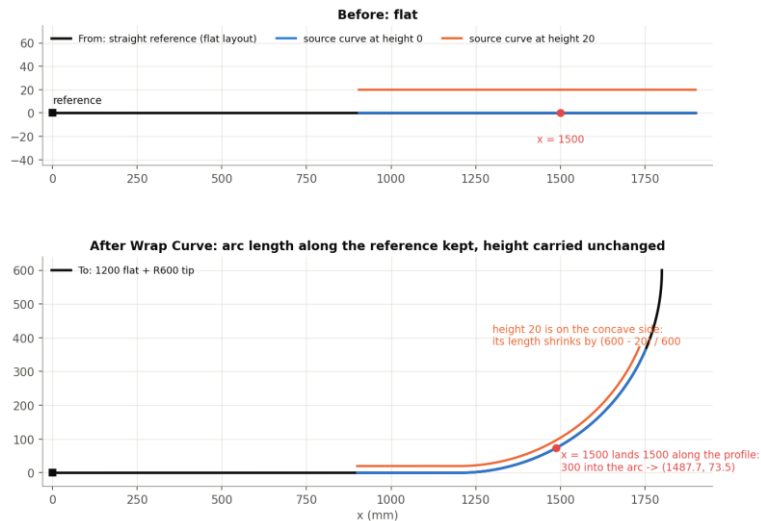
Deform: every vertex, edge and face goes through the same map



A 100 x 5 block bent onto the tip: bottom 100, top 99.2 (concave side).



The bending map



Arc length kept

A point s along the From reference lands s along the To reference (a shift, no scaling).

Height and width kept

The point's offsets in the frame carry over: ribs stay perpendicular.

Layers bend

At height h around radius R they stretch or shrink by $(R - h) / R$, as a real layer does.



The dialog

IMAGE MISSING:
docs/decks/deform/shots/dialog.png

#	Parameter	What it means
1	Mode	Bodies (one solid or sheet) / Faces.
2	From data	From edge(s) (G1 chain) and From reference (vertex, mate connector, plane).
3	To data	To edge(s), Flip (reverse To), To reference, Flip normal.
4	Frame orientation	Frenet (default) / Binormal (face, mate connector or world axis).
5	Setup	Sampling, degree, max control points, tolerance; Use face guide points; Keep wires.
6	Debug	Create wires / faces / bodies stages; show frames.



Outputs and messages

Outputs

Key	What it holds
(body)	A new body (the source stays); solids closed into solids.

Messages

	Message / what to do
Console	per-face WARNING / ERROR, OPEN END reports Check for a missing face or a sheet instead of a solid.



Tips and limits



Every face is a fill

Planes and cylinders are not kept as such; inner loops may fail.



Guide points

Turn on Use face guide points when fills do not follow the face.



Heavy

The slowest of the family: many edges, many projections.



Where it lives

Document	curveMapping_public (d12583277c42b00c63a9b7b7)
Tab	deform (+ engine curveMappingCore)
Tests	none
Explainer	docs/explainers/curve_mapping/curve_mapping_explained.md, section 2.3
Related	Wrap Curve, Wrap and Loft, Deform, Measure curve distance, Offset edges